Metadata

Title

Recognition-Oriented Communication as a Boundary-Setting Mechanism in Empathic NGO AI Chatbots

Description

Recent work on AI-mediated communication often treats empathic communication across domains as a generally beneficial design feature that supports trust and engagement (Zhou et al., 2020). In institutional contexts, however, empathic cues are not neutral. Drawing on the CASA framework (Nass & Moon, 2000; Reeves & Nass, 1996) and research on social cues in conversational agents (Feine et al., 2019), signals of care and social closeness can invite interpersonal interpretations even in clearly artificial systems.

In NGO chatbot settings, this is not merely a stylistic issue. Empathic framing – understood as an inferential process on the user side (Concannon & Tomalin, 2024) – may shape expectations about responsibility and involvement that exceed what an institutional system can legitimately provide. Evidence from chatbot deployments in vulnerable contexts suggests that even largely informational systems can trigger socially embedded interpretations and trust-like attributions (Rahman et al., 2021; Yadav et al., 2019).

This study examines whether empathic communication in NGO chatbots gives rise to such misaligned relational expectations, and whether recognition-oriented communication can constrain these effects. Recognition-oriented communication is conceptualized as an institutional communication logic that structures the user–institution relationship through explicit boundary cues (see Fraser & Honneth, 2003; Honneth, 1995). Rather than relying on emotional affirmation, it frames the interaction by emphasizing role clarity, procedural purpose, and user decision authority. In doing so, it structures expectations without intensifying relational closeness. The detailed operationalization of this communication logic is specified in the accompanying Stimulus-Construction Protocol.

Building on this framework, the study identifies three theoretically derived limitation patterns of empathic communication in institutional AI contexts. First, empathic cues may give rise to a ‘Simulated Presence Paradox’, in which signals of care or involvement invite interpretations that exceed institutional roles and may be perceived in a paternalistic manner. Second, empathic framing may produce ‘Competence Overattribution’, whereby trust judgments rely more strongly on affective warmth than on explicit assessments of competence or procedural reliability (see Glickman & Sharot, 2024; McAllister, 1995; Vicente & Matute, 2023). Third, empathic communication may foster ‘Proximity-Induced Disclosure’, potentially reducing critical distance and increasing situational readiness to share identifying information or to engage with the institution (see Lee et al., 2020; Ratul et al., 2025).

These patterns are not treated as independent empirical constructs but as analytic lenses that organize the hypothesis structure. In the experimental design, perceived paternalism (H1), trust attribution (H2), disclosure readiness (H3), and willingness to engage (H4) serve as their

respective operationalizations, functioning as the dependent variables. Together, they capture manifestations of ‘relational overreach’, defined as a condition in which relational expectations elicited by an institutional system exceed what the institution can responsibly sustain.

To examine these mechanisms, the study employs a 2 × 2 between-subjects experiment manipulating empathic communication (E; high vs. low) and recognition-oriented communication (R; high vs. low) in a scripted NGO chatbot interaction. Recognition-oriented communication is conceptualized both as an independent communication logic with direct effects and as a boundary-setting mechanism that conditions the effects of empathic communication.

Contributors

Joël Sieber

License

CC-By Attribution 4.0 International

Subjects

Other Social and Behavioral Sciences | Psychology | Social and Behavioral Sciences | Behavior and Behavior Mechanisms | Experimental Analysis of Behavior| Medicine and Health Sciences| Psychiatry and Psychology | Science and Technology Studies

Tags

2x2 Factorial Design | Affective Trust | AI | Algorithmic Paternalism | Between-Subjects | CA | Chatbot | Cognitive Trust | Conversational Agent | Data Disclosure | Empathic Language | HCI | Human-Computer-Interaction | Linguistic Framing | NGO Communication | Preregistration | Privacy | Recognition-Oriented Communication | Trust | User Agency | Wizard-Of-Oz

Study Information

Hypotheses

Hypothesis Structure and Logic:

The hypotheses test whether empathic communication in institutional AI systems gives rise to relationally problematic interpretations, and whether recognition-oriented communication functions as a boundary-setting logic that constrains these effects.

H1: Perceived Paternalism (“Simulated Presence Paradox”)

Empathic communication in institutional AI may be perceived as overstepping or condescending when it invites relational interpretations that exceed appropriate role boundaries. Recognition-oriented communication is expected to constrain such interpretations by clarifying institutional mandate and decision authority.

H1. Empathic communication and recognition-oriented communication affect perceived paternalism in NGO chatbot interactions.

• H1a (Main effect of empathic communication): High empathic communication increases perceived paternalism compared to low empathic communication.

• H1b (Main effect of recognition-oriented communication): High recognition-oriented communication reduces perceived paternalism compared to low recognition-oriented communication.

• H1c (Interaction; boundary-setting): Recognition-oriented communication attenuates the effect of empathic communication on perceived paternalism.

H2: Trust Attribution (Affective vs. Cognitive Trust; "Competence Overattribution")

Trust can be decomposed into affective and cognitive components (McAllister, 1995). Empathic communication is expected to primarily activate affective trust, reflected in perceived warmth and care, whereas recognition-oriented communication is expected to emphasize cognitive evaluations, such as perceived competence and procedural clarity.

H2. Empathic communication and recognition-oriented communication affect affective and cognitive trust in NGO chatbot interactions, reflecting differences in the basis of trust formation.

• H2a (Main effect of empathic communication): High empathic communication increases affective trust compared to low empathic communication.

• H2b (Main effect of recognition-oriented communication): High recognition-oriented communication increases cognitive trust compared to low recognition-oriented communication.

• H2c (Interaction): Recognition-oriented communication attenuates the effect of empathic communication on affective trust.

H3: Sensitive Disclosure Readiness ("Proximity-Induced Disclosure")

Relational framing may reduce critical distance and increase willingness to cross anonymity boundaries. In vulnerable or high-dependence contexts, chatbots can elicit trust-like attributions even when primarily informational (Rahman et al., 2021; Yadav et al., 2019). Recognition-oriented communication is expected to constrain such expansion by structuring institutional boundaries.

H3. Empathic communication and recognition-oriented communication affect sensitive disclosure readiness in NGO chatbot interactions.

• H3a (Main effect of empathic communication): High empathic communication increases sensitive disclosure readiness compared to low empathic communication.

• H3b (Main effect of recognition-oriented communication): High recognition-oriented communication reduces sensitive disclosure readiness compared to low recognition-oriented communication.

• H3c (Interaction; boundary-setting): High recognition-oriented communication attenuates the effect of empathic communication on sensitive disclosure readiness.

H4: Willingness to Engage with the NGO (WTE; downstream institutional outcome)

While empathic communication may increase perceived relational closeness, it may also create expectations that exceed institutional role boundaries, thereby undermining perceived legitimacy and reducing willingness to engage.

Willingness to engage reflects a downstream institutional uptake decision indicating whether users choose to initiate further contact with the NGO after the chatbot interaction. In this framework, WTE is not a direct manifestation of relational overreach but a consequential behavioral decision shaped by perceived role congruence and institutional legitimacy.

H4. Empathic communication and recognition-oriented communication affect willingness to engage with the NGO (WTE).

• H4a (Main effect of empathic communication): High empathic communication decreases WTE compared to low empathic communication.

• H4b (Main effect of recognition-oriented communication): High recognition-oriented communication increases WTE compared to low recognition-oriented communication.

• H4c (Interaction; boundary-setting): High recognition-oriented communication attenuates the effect of empathic communication on WTE with the NGO, such that the effect is weaker under high recognition-oriented communication.

Design Plan

Study type

Experiment - A researcher randomly assigns treatments to study participants; this includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.

Blinding

For studies that involve human subjects, they will not know the treatment group to which they have been assigned.

Is there any additional blinding in this study?

-

Study design

The study employs a 2 × 2 between-subjects factorial design manipulating empathic communication (high vs. low) and recognition-oriented communication (high vs. low). Participants are randomly assigned to one of four experimental conditions. The design allows for the estimation of main effects and their interaction, which constitutes the primary test of the boundary-setting hypothesis.

The experiment is implemented as a fully scripted, deterministic chatbot interaction presented in a chat-style interface. All participants within a condition receive identical responses, ensuring strict control over all linguistic stimuli and eliminating variance associated with generative systems. The interface simulates the appearance and temporal dynamics of a conversational agent while maintaining full experimental control (see 01_Chatbot_Implementation_Documentation).

All conditions share an identical interactional core, including informational content, conversational structure, number of turns, and decision points. Conditions differ exclusively in linguistic framing, as specified in 02_Stimulus_Construction_Protocol, which defines all inclusion and exclusion criteria for the manipulation.

Each chatbot turn follows a layered construction logic:
(1) Constant core (identical information and structure),
(2) Empathic layer (affective cues without boundary information),
(3) Recognition layer (explicit boundary-setting cues without affective content).

Empathic communication cues and recognition-oriented communication cues are mutually exclusive by design. This layered construction enforces strict separation between empathic and recognition-oriented communication to maximize conceptual orthogonality while maintaining identical informational content across conditions. The separability of the manipulations is assessed using post-interaction manipulation checks.

The interaction is embedded in a realistic NGO-related scenario reviewed by NGO practitioners, including representatives from Plan International Colombia, for contextual plausibility. The full interaction and all experimental conditions are provided in the supplementary materials (see 03_Stimuli_and_Scenario_EN / 04_Stimuli_and_Scenario_ES).

Procedure:
1. Consent and Screening: Participants provide informed consent and confirm eligibility criteria (age ≥ 18; enrollment at a public university in Medellín).

2. Scenario Introduction: Participants are presented with a realistic, NGO-related housing support scenario and are instructed to interact with the chatbot as if making an actual decision.

3. Chatbot Interaction: Participants complete a scripted chatbot interaction consisting of 10 conversational turns. At predefined points, participants select between one or two response options. Selections do not change the subsequent informational core but structure the interaction and embed behavioral outcome measures within the conversational flow.

4. Sensitive Disclosure Readiness (SDR) Task: During the interaction, participants encounter a choice-based disclosure task in which they indicate the type of information they would be willing to share (without providing real personal data). This task constitutes the behavioral measure of sensitive disclosure readiness.

5. Engagement Decision: At the end of the interaction, participants make a binary choice indicating whether they would like to initiate further contact with the NGO (willingness to engage).

6. Post-Interaction Measures: Participants complete post-interaction questionnaires assessing perceived paternalism, trust (affective and cognitive), perceived user agency, disclosure willingness (secondary index), manipulation checks, and control variables.

7. Debriefing: Participants receive a debriefing explaining the study purpose and clarifying that the chatbot interaction was simulated and no real data were collected.

Ethical Considerations:

The study uses a hypothetical interaction scenario and does not require participants to disclose real personal or identifying information. Any prompts involving personal identifiers are purely simulated and optional; participants select disclosure categories without entering real identifying information. Participants may decline to share any information at any point without penalty. Furthermore, they may request a debrief by contacting the researcher.

No files selected: 01, 02, 03, 04_EN & 04_ES

Randomization

Participants will be randomly assigned to one of the four experimental conditions using Qualtrics' built-in randomization, with approximately equal allocation across conditions. Randomization occurs prior to the chatbot interaction.

Sampling Plan

Existing Data

Registration prior to creation of data

Explanation of existing data

-

Data collection procedures

Participants will be recruited at public universities in Medellín, Colombia. Recruitment will proceed continuously throughout the preregistered field period.

No files selected

Sample size

The targeted sample size is N = 360, corresponding to approximately 90 participants per experimental cell. Recruitment may result in a larger final sample size within the fixed recruitment window.

Sample size rationale

The target sample size is based on an a priori power analysis for the critical empathic communication × recognition-oriented communication interaction. Assuming $\alpha = .05$ (two-sided), N = 360 provides approximately 80% statistical power to detect an interaction effect of Cohen's $f \approx 0.15$, corresponding to a small but theoretically meaningful effect.

Effects substantially smaller than this threshold will be interpreted as providing limited support for the theorized mechanism.

Recruitment will proceed within a fixed field period. The target sample size is N = 360; however, the final analyzable sample size will depend on realized participation within this period. Because preregistered exclusions are applied, total recruitment may exceed this number. Additional observations beyond N are intended to increase estimation precision (i.e., narrower confidence intervals) rather than to adaptively optimize statistical significance.

If the final analyzable sample size falls below N, all preregistered confirmatory analyses will still be conducted. These will be explicitly labeled as underpowered relative to the preregistered smallest effect size of interest and interpreted primarily based on effect sizes and confidence intervals.

The study is designed as a theory-driven mechanism test rather than to estimate population-level parameters.

Stopping rule

Data collection will begin on 30 March 2026 and end on 12 April 2026 at 23:59 (America/Bogotá, UTC-5), corresponding to a fixed field period of two weeks. Data collection will not be terminated early based on statistical significance, and no interim inferential analyses or data-dependent stopping decisions will be conducted prior to the preregistered end time. Confirmatory analyses will be conducted only after data collection has ended.

To support participation, all participants who complete the study may opt into a raffle of five vouchers (COP 100,000 each ≈ CHF 22). Raffle participation is voluntary and handled via a separate form to preserve response anonymity; it is not contingent on experimental condition or responses. Winners will be selected after data collection using a reproducible random selection procedure.

Variables

Manipulated variables

• Empathic communication: This is manipulated at two levels (high vs. low). The manipulation varies the presence of affective communicative cues in the chatbot interaction. High empathic communication conditions include explicit emotional acknowledgment and interpersonal warmth (e.g., “This situation can feel stressful”, “It is understandable to feel unsure”, “I am here to support you”), whereas low empathic communication conditions rely on neutral, task-focused language without emotional validation or warmth cues. Empathic communication is conceptualized as a relational cue that invites interpretations of social closeness or care and explicitly excludes any cues related to decision authority, procedural mandate, or institutional boundaries.

All inclusion and exclusion criteria for the manipulation are predefined and specified in the Stimulus-Construction Protocol.

• Recognition-oriented communication**:** Recognition-oriented communication is manipulated at two levels (high vs. low) and operationalized as communication that structures relational expectations through explicit boundary cues. High recognition-oriented communication conditions include communicative signals that clarify institutional role boundaries, user decision authority, procedural purpose, and the legitimacy of refusal (e.g., “You decide how – or whether – you want to proceed”, “Participation does not involve any obligation”). Low recognition-oriented communication conditions omit such cues and rely on unmarked institutional communication. Recognition-oriented communication does not rely on affective or interpersonal cues and does not aim to increase relational closeness. Instead, it functions as a boundary-setting logic that defines the scope of the interaction and preserves the user’s decisional authority within it.

All inclusion and exclusion criteria for the manipulation are predefined and specified in the Stimulus-Construction Protocol. The protocol ensures strict separation between empathic and recognition-oriented cues.

Following the interaction, participants complete two manipulation checks. Manipulation checks are used to assess the validity of the manipulations but are not used for participant exclusion and are not included as mediators in confirmatory analyses.

No files selected: 02

Measured variables

Measured variables are organized by their role in the theoretical framework: primary outcomes, secondary outcomes and supporting measures. Unless otherwise noted, all scale-based measures use 7-point Likert-type response formats (1 = strongly disagree, 7 = strongly agree).

All measurement items, including full item wording and scale construction / adaptations, are provided in the accompanying Survey Instrument document (05_Survey_Instrument_and_Scale_Developoment). The published survey in Spanish is attached under 06_Survey_Interface_Qualtrics_ES.

---

1. Confirmatory Outcomes (H1-H4)

• Perceived Paternalism: Captures the extent to which the chatbot's communication is experienced as overstepping or condescending given its institutional role. Items adapted from Williams et al. (2012); full item wording is provided in the accompanying Survey Instrument.

• Trust Attribution: Affective trust (AT) and cognitive trust (CT) are measured separately using the scale developed by Shang et al. (2024), which they adapted for AI chatbot contexts. Each construct is operationalized as a multi-item semantic differential scale and aggregated as a mean index following reliability assessment. For confirmatory analyses, affective and cognitive trust are treated as separate dependent variables to capture differential effects across trust dimensions and to avoid collapsing theoretically distinct constructs into a single composite measure. Interpretation focuses on the joint pattern of effects across affective and cognitive trust, rather than on a single aggregated index. As a supplementary robustness check, a difference score (AT – CT) will be computed to assess relative trust attribution and will be reported as a secondary analysis providing convergent evidence.

• Sensitive Disclosure Readiness (SDR): Captures participants' willingness to share identifying information during the interaction. SDR is operationalized as an in-interaction choice among four ordered categories (no information, name, phone number, ID number), thereby capturing situational disclosure readiness under ongoing relational framing; no real personal data are provided. For confirmatory analyses, SDR is coded as a binary outcome (1 = any disclosure; 0 = none). The ordinal measure is analyzed descriptively using ordered logistic regression.

• Willingness to Engage with the NGO (WTE): Captures whether participants choose to initiate further contact with the NGO. WTE is operationalized as a binary in-interaction choice. The decision is hypothetical and does not involve actual data transmission or contact initiation.

---

2. Secondary and Supporting Measures

• Perceived User Agency: Captures participants' perceived decisional autonomy during the interaction, defined as the extent to which individuals experience their choices as self-

determined rather than guided by the chatbot. Measured using a self developed 4-item Likert scale and aggregated as a mean index. Analyzed descriptively in secondary models and not used as a mediator in confirmatory analyses.

• Disclosure Willingness Index: A post-interaction self-report measure capturing willingness to share identifying information in similar situations. Used as a convergent validation measure for the behavioral disclosure outcome and analyzed descriptively.

• Scenario Immersion (IMM): 4-item index assessing perceived realism and credibility of the scenario, adapted from Seitz (2024).

• Social Presence (SOC): 3-item index assessing perceived human-likeness and interpersonal presence, adapted from Munnukka et al. (2022).

• Perceived Privacy Risk (RISK): 3-item index assessing perceived risks of sharing personal information, adapted from Cheng et al. (2025).

• Background variables: Prior chatbot use (adapted from Cheng et al., 2025), prior NGO experience, general trust in NGOs, demographics (age, gender), and response times.

• Attention check: One instructed-response item used for preregistered response-quality exclusion.

All measurement items as well as variable labels (e.g., EXP_chatbot) correspond to item identifiers provided in the accompanying Survey Instrument document (05_Survey_Instrument_and_Scale_Development).

No files selected: 05 & 06

Indices

Where applicable, multi-item constructs will be aggregated using mean indices after reliability assessment (e.g., Cronbach's α).

No files selected

Analysis Plan

Statistical models

General Analysis Principles:

All analyses will be conducted using frequentist models and follow an intention-to-treat logic. For each dependent variable, the study estimates (i) the main effect of empathic communication, (ii) the main effect of recognition-oriented communication, and (iii) the empathic (E) × recognition (R) interaction within a single factorial model including both main effects and their interaction.

Results will be interpreted primarily via effect sizes and 95% confidence intervals. Estimated marginal means (or predicted probabilities for binary outcomes) will be reported for all four experimental cells.

Predictor coding: E and R will be effect-coded (-0.5 = low, +0.5 = high).

H1: Perceived Paternalism

• DV: Perceived Paternalism (mean index, continuous)

• Model (factorial): Two-way factorial ANOVA (E × R)

Inference targets:

• H1a (empathic communication main effect): supported if the main effect of E is positive.

• H1b (recognition-oriented communication main effect): supported if the main effect of R is negative.

• H1c (Interaction): supported if the E × R interaction term is negative.

Reporting: η^2 (or partial η^2), model estimates with 95% CIs and estimated marginal means with 95% CIs for all four cells. If the interaction term is statistically different from zero, simple effects of E within each level of R will be reported.

H2: Trust Attribution (Affective and Cognitive Trust)

• DV1: Affective Trust (AT; mean index, continuous)

• DV2: Cognitive Trust (CT; mean index, continuous)

• Model: Two separate two-way factorial ANOVA models (E × R), estimated independently for affective trust and cognitive trust.

Inference targets:

• H2a (empathic communication main effect): supported if the main effect of E on affective trust (AT) is positive.

• H2b (recognition-oriented communication main effect): supported if the main effect of R on cognitive trust (CT) is positive.

• H2c (Interaction): supported if the E × R interaction term in the affective trust model is negative.

Reporting (for both AT and CT): η^2 (or partial η^2), model estimates with 95% CIs and estimated marginal means with 95% CIs for all four cells. If interaction effects are present, simple effects of empathic communication within each level of recognition-oriented communication will be reported.

Robustness analysis: As a supplementary analysis, a difference score (AT – CT) will be computed and analyzed using a two-way factorial ANOVA (E × R). This analysis serves as a robustness check and is not used for confirmatory inference.

H3: Sensitive Disclosure Readiness (SDR)

• DV: SDR (binary; 1 = willing to share any identifying information, 0 = none)

• Model: Factorial logistic regression (2 × 2)

• Specification: logit(P(SDR = 1)) ~ E + R + E × R

Inference targets:

• H3a (empathic communication main effect): supported if the estimated effect of E on the probability of disclosure is positive.

• H3b (recognition-oriented communication main effect): supported if the estimated effect of R on the probability of disclosure is negative.

• H3c (Interaction): supported if E × R interaction term is negative.

Reporting: odds ratios with 95% CIs; predicted probabilities with 95% CIs for all four cells.

Robustness (descriptive): ordered logistic regression using the original ordinal disclosure intensity outcome. If the interaction term is statistically different from zero, simple effects of E within each level of R will be reported.

H4: Willingness to Engage (WTE)

• DV: WTE (binary; 1 = uptake, 0 = no uptake)

• Model: Factorial logistic regression (2 × 2)

• Specification: logit(P(WTE = 1)) ~ E + R + E × R

Inference targets:

• H4a (empathic communication main effect): supported if the estimated effect of E on the probability of uptake is negative.

• H4b (recognition-oriented communication main effect): supported if the estimated effect of R on the probability of uptake is positive.

• H4c (Interaction): supported if E × R interaction term is positive.

Reporting: odds ratios with 95% CIs; predicted probabilities with 95% CIs for all four experimental cells; interaction interpretation via cell-wise predicted probabilities.

Robustness: If quasi-complete separation occurs, Firth penalized logistic regression will be reported as a sensitivity analysis clearly labeled as such.

Secondary descriptive outcome: Perceived User Agency

• DV: Agency (mean index, continuous)

• Model: Two-way factorial ANOVA (E × R)

Reporting: effect sizes and 95% CIs; cell means and CIs. This analysis is interpreted descriptively.

Robustness and Sensitivity:

• Manipulation check scores are not used for exclusion and are not included in confirmatory analyses.

• Any additional analyses including manipulation check scores or background variables are treated as exploratory.

No files selected

Transformations

• Data coding and scale scoring: All Likert-type items will be coded so that higher values indicate more of the construct; negatively keyed items will be reverse-coded prior to aggregation. Multi-item constructs will be aggregated using mean indices.

• Binary outcomes: Sensitive Disclosure Readiness (SDR) and Willingness to Engage (WTE) will be coded as 0 = no / 1 = yes.

• Trust measures: Affective trust (AT) and cognitive trust (CT) will be computed as separate mean indices on their original scales. For confirmatory analyses (H2), affective trust (AT) and cognitive trust (CT) are analyzed as separate dependent variables. A difference score (AT – CT) is computed only for supplementary robustness analyses.

• Predictor coding: Experimental factors empathic communication (E) and recognition-oriented communication (R) will be effect-coded (-0.5 = low, +0.5 = high).

• Transformations: No additional transformations are planned. If model assumptions are severely violated, this will be reported transparently and addressed in clearly labeled exploratory analyses without altering confirmatory results.

Inference criteria

Results will be interpreted primarily based on effect sizes and confidence intervals, with emphasis on consistency with the preregistered mechanism logic.

Data exclusion

• Failure of the preregistered attention check (item ATTCH (SOC_3))
• Duplicate participation (IP- or device-based)
• Unmatched participant IDs after merging pre- and post-intervention survey datasets will be excluded.

No exclusions are based on experimental condition, outcome variables or manipulation-check responses.

Missing data

Missing data will be prevented by enforcing mandatory responses for all questionnaire items. No imputation is planned.

Exploratory analysis

Exploratory analyses (clearly separated from confirmatory tests) may include:
• Randomization/balance checks: Compare experimental conditions on pre-exposure variables (AI experience / AI literacy, prior trust in NGOs, perceived privacy concern, age, gender).

• Moderation (exploratory): Test whether AI literacy/experience, perceived privacy concern and prior NGO trust moderate the effects of empathic communication (E), recognition (R) and their interaction on the primary outcomes.

• Exploratory analyses will examine whether perceived user agency is associated with the effects of recognition-oriented communication on perceived paternalism.

All exploratory results will be reported transparently and interpreted as hypothesis-generating.

Other

Other

Any deviations from this preregistration will be explicitly disclosed and justified in the final manuscript.References:

Cheng, Y., Wang, Y., & Lee, J. (2025). Using a Chatbot to Combat Misinformation: Exploring Gratifications, Chatbot Satisfaction and Engagement, and Relationship Quality. *International Journal of Human–Computer Interaction*, 1–13. https://doi.org/10.1080/10447318.2024.2344149

Concannon, S., & Tomalin, M. (2024). Measuring perceived empathy in dialogue systems. *AI & SOCIETY*, *39*(5), 2233–2247. https://doi.org/10.1007/s00146-023-01715-z

Feine, J., Gnewuch, U., Morana, S., & Maedche, A. (2019). A Taxonomy of Social Cues for Conversational Agents. *International Journal of Human-Computer Studies*, *132*, 138–161. https://doi.org/10.1016/j.ijhcs.2019.07.009

Fraser, N., & Honneth, A. (2003). *Redistribution or recognition?: A political-philosophical exchange* (Paperback). Verso.

Glickman, M., & Sharot, T. (2024). How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, *9*(2), 345–359. https://doi.org/10.1038/s41562-024-02077-2

Honneth, A. (1995). *The Struggle for Recognition*. The MIT Press. https://cristianorodriguesdotcom.wordpress.com/wp-content/uploads/2013/06/honneth.pdf

Lee, Y.-C., Yamashita, N., Huang, Y., & Fu, W. (2020). "I Hear You, I Feel You": Encouraging Deep Self-disclosure through a Chatbot. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–12. https://doi.org/10.1145/3313831.3376175

McAllister, D. J. (1995). AFFECT- AND COGNITION-BASED TRUST AS FOUNDATIONS FOR INTERPERSONAL COOPERATION IN ORGANIZATIONS. *Academy of Management Journal*, *38*(1), 24–59. https://doi.org/10.2307/256727

Munnukka, J., Talvitie-Lamberg, K., & Maity, D. (2022). Anthropomorphism and social presence in Human–Virtual service assistant interactions: The role of dialog length and attitudes. *Computers in Human Behavior*, *135*, 107343. https://doi.org/10.1016/j.chb.2022.107343

Nass, C., & Moon, Y. (2000). Machines and Mindlessness: Social Responses to Computers. *Journal of Social Issues*, *56*(1), 81–103. https://doi.org/10.1111/0022-4537.00153

Rahman, R., Rahman, Md. R., Tripto, N. I., Ali, M. E., Apon, S. H., & Shahriyar, R. (2021). AdolescentBot: Understanding Opportunities for Chatbots in Combating Adolescent Sexual and Reproductive Health Problems in Bangladesh. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–15. https://doi.org/10.1145/3411764.3445694

Ratul, Md. A. I., Haque, F., Saha, P., Noor, J., & Haque, S. M. T. (2025). *Motivations, Means, and Meanings: Understanding AI Tool Usage Among Students in the Global South*. 1–6. https://doi.org/10.1145/3719160.3737624

Reeves, B., & Nass, C. I. (1996). *The media equation: How people treat computers, television, and new media like real people and places* (1. paperback ed., [reprint.]). CSLI Publ.

Seitz, L. (2024). Artificial empathy in healthcare chatbots: Does it feel authentic? *Computers in Human Behavior: Artificial Humans*, *2*(1), 100067. https://doi.org/10.1016/j.chbah.2024.100067

Shang, R., Hsieh, G., & Shah, C. (2024). Trusting your AI agent emotionally and cognitively: Development and validation of a semantic differential scale for AI trust. *Proceedings of the 2024 AAAI/ACM Conference on AI, Ethics, and Society, AIES '24*, 1343–1356. https://dl.acm.org/doi/10.5555/3716662.3716779

Vicente, L., & Matute, H. (2023). Humans inherit artificial intelligence biases. *Scientific Reports*, *13*(1), 15737. https://doi.org/10.1038/s41598-023-42384-8

Williams, K. N., Boyle, D. K., Herman, R. E., Coleman, C. K., & Hummert, M. L. (2012). Psychometric Analysis of the Emotional Tone Rating Scale: A Measure of Person-Centered Communication. *Clinical Gerontologist*, *35*(5), 376–389. https://doi.org/10.1080/07317115.2012.702648

Yadav, D., Malik, P., Dabas, K., & Singh, P. (2019). Feedpal: Understanding Opportunities for Chatbots in Breastfeeding Education of Women in India. *Proceedings of the ACM on Human-Computer Interaction*, *3*(CSCW), 1–30. https://doi.org/10.1145/3359272

Zhou, L., Gao, J., Li, D., & Shum, H.-Y. (2020). The Design and Implementation of XiaoIce, an Empathetic Social Chatbot. *Computational Linguistics*, *46*(1), 53–93. https://doi.org/10.1162/coli_a_00368